Abstract

We consider conditions under which a universe contracting towards a big crunch can make a transition to an expanding big bang universe. A promising example is 11-dimensional M theory in which the eleventh dimension collapses, bounces, and reexpands.

At the bounce, the model can reduce to a weakly coupled heterotic string theory and, we conjecture, it may be possible to follow the transition from contraction to expansion. The possibility opens the door to new classes of cosmological models.

For example, we discuss how it suggests a major simplification and modification of the recently proposed ekpyrotic scenario.

Keywords: Big Bang, Big Crunch, 11th Dimension, Heterotic String Theory, Cosmological Models.